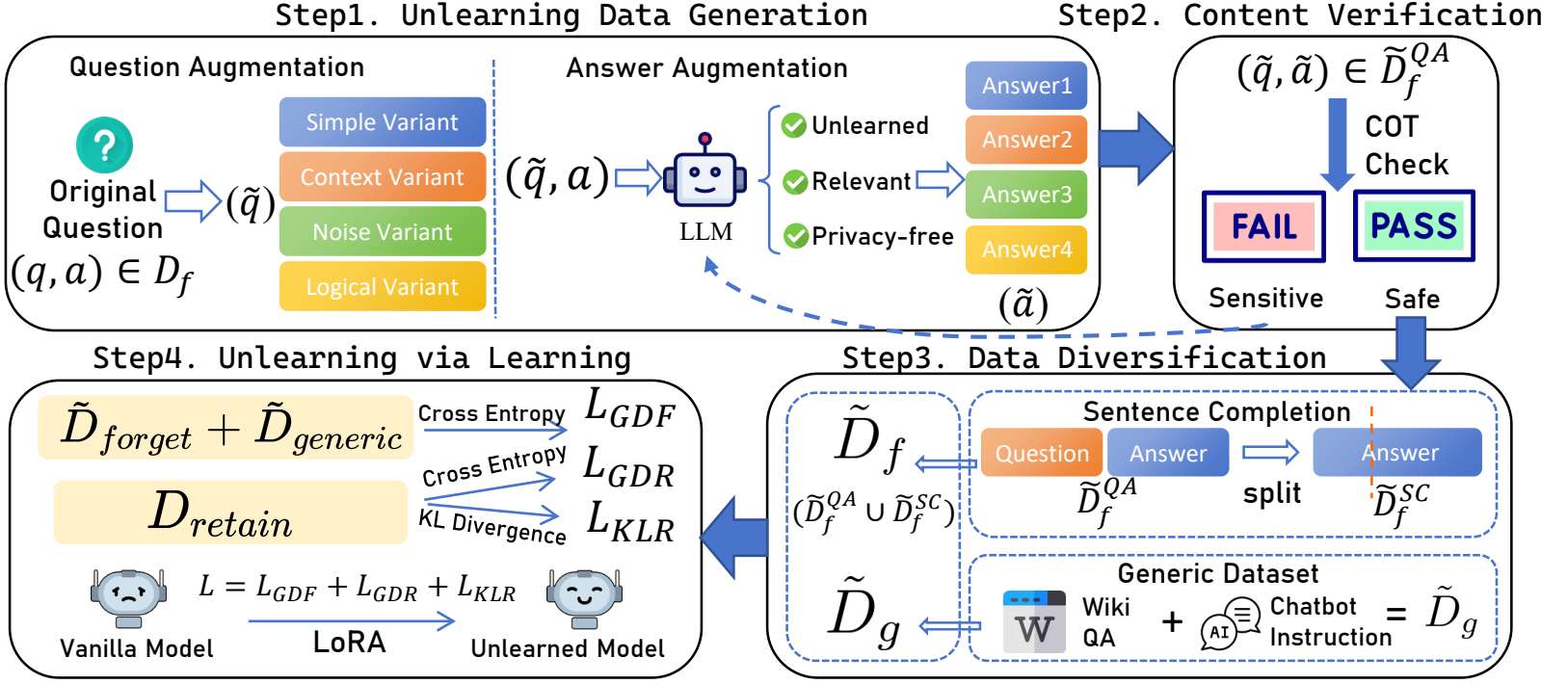
Step1. Unlearning Data Generation
Question Augmentation
Original Question
$(q, a) \in D_f$
$(\tilde{q})$
Simple Variant
Context Variant
Noise Variant
Logical Variant
Answer Augmentation
$(\tilde{q}, a)$
LLM
Unlearned
Relevant
Privacy-free
Answer1
Answer2
Answer3
Answer4
$(\tilde{a})$
Step2. Content Verification
$(\tilde{q}, \tilde{a}) \in \tilde{D}_f^{QA}$
COT Check
FAIL
PASS
Sensitive
Safe
Step3. Data Diversification
$\tilde{D}_f$
$(\tilde{D}_f^{QA} \cup \tilde{D}_f^{SC})$
Sentence Completion
Question
Answer
split
Answer
$\tilde{D}_f^{QA}$
$\tilde{D}_f^{SC}$
$\tilde{D}_g$
Generic Dataset
Wiki QA
+
Chatbot Instruction
$= \tilde{D}_g$
Step4. Unlearning via Learning
$\tilde{D}_{forget} + \tilde{D}_{generic}$
D_{retain}
Cross Entropy
Cross Entropy
KL Divergence
L_{GDF}
L_{GDR}
L_{KLR}
$L = L_{GDF} + L_{GDR} + L_{KLR}$
LoRA
Vanilla Model
Unlearned Model